IEEE Access
Multidisciplinary ⋮ Rapid Review ⋮ Open Access Journal

Received 2 October 2025, accepted 11 October 2025, date of publication 23 October 2025, date of current version 6 November 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3624766

RESEARCH ARTICLE

A Game Theory-Based Approach Considering Car-Following Behavior for Modeling Merging Behavior at Expressway On-Ramp Sections

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ABSTRACT Merging at expressway on-ramp sections is a typical mandatory lane-changing maneuver that often causes traffic conflicts and bottlenecks. Although game-theoretic models have been widely applied to describe merging interactions, many existing studies pay little attention to the car-following behavior of the lag vehicle, particularly at taper-type merging sections where drivers have limited space and time to react. This study develops a game-theory-based merging model that explicitly incorporates car-following dynamics into drivers' decision making. The merging process is formulated as a non-zero-sum, non-cooperative game between the merging vehicle and the lag vehicle, whose strategies are defined as merge/wait and yield/keep-following/block, respectively. A rule-based identification method was proposed to extract observed vehicle actions from trajectory data, and the model was calibrated and validated using field observations from Hanshin Expressway. The results demonstrate that the proposed model achieved a 5.2% improvement in overall prediction accuracy compared with representative game-theoretic models, and it also provided consistent trajectory prediction through the integrated car-following formulation. The main contribution of this study is: i) it extends game-theoretic merging models by considering LV's continuous car-following behavior, thereby capturing more realistic human driving decisions; and (ii) it offers a validated modeling framework applicable to taper-type merging sections, providing a foundation for microscopic traffic simulation and automated vehicle control strategies.

INDEX TERMS Game theory, merging behavior, car-following model.

I. INTRODUCTION

Lane-changing behavior is an important topic in traffic flow modeling. Drivers need to select an appropriate gap and interact with surrounding vehicles to make lane-changing decisions. As a result of such interactions, conflicts may occur between vehicles and create bottlenecks in congested traffic. Many studies have emphasized the significant impact of lane-changing maneuvers on traffic flow stability and capacity. For example, Laval and Daganzo [1] demonstrated that lane changes can trigger stop-and-go waves in dense freeway traffic, while Zheng [2] provided a comprehensive review of lane-changing models highlighting their influence on traffic dynamics.

Game theory is a powerful tool for the study of interaction among human drivers and their decision-making. Game theory-based lane-changing models have been studied for decades, but a few issues remain to be addressed. Schakel et al. [3] integrated a gap acceptance-based lane-changing model with a modified car-following model to study the behavior of the subject vehicle in lane changing. A few studies have integrated game theory-based models with car-following models to reflect the behavior of both subject vehicles and surrounding vehicles. Recent studies such as Liao et al. [4] and Min et al. [5] combined game theory-based decision making and acceleration models of autonomous vehicle merging control. But the car-following

The associate editor coordinating the review of this manuscript and approving it for publication was Tiago Cruz.

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behavior of human drivers during merging interaction still remains an issue. Ji and Levinson [6] also pointed out that game theory-based lane changing models will be modified to cooperate with (or incorporate) some of the well-known car-following models or directly adapt to traffic simulation tools.

Besides, the American Association of State Highway and Transportation Officials (AASHTO) [7] recommends a 360 m parallel acceleration lane that includes a 90 m taper section for the expressway on-ramp section in the guidelines. Previous studies mainly set the scenario on parallel-type merging sections [8], [9]. However, in many cases, taper-type, on-ramp sections without parallel acceleration lanes are common [10]. At the taper-type merging section, drivers have fewer opportunities and less reaction time to select an acceptable gap due to the limited length of taper; therefore, the interaction between ramp and mainline vehicles is different from the parallel-type cases.

The objective of this study was to develop a game theory-based, on-ramp-merging, lane-changing model incorporating car-following models, especially the mainline lag vehicle's car-following behaviors, and applicable to taper-type merging sections. The main contributions of this study are as follows:

(i) A game-theory based model was formulated considering car-following behavior of mainline vehicles, and car-following models for different vehicle strategies were developed respectively.

(ii) The proposed model was calibrated and validated utilizing the real-world observation data of a taper-type, on-ramp-merging section on Hanshin Expressway in Japan. Results demonstrated the soundness and consistency of the model.

The rest of the paper is organized as follows: Section II reviews previous studies on game theory-based lane-changing models. Section III presents the methodology, including the formulation of merging game and definition of car-following model. Section IV describes the data source, data pre-processing and empirical evidence of strategies based on the proposed identification method. Section V describes the model calibration and validation results and makes a comparison with previous studies. Section VI summarizes the main findings and issues.

II. LITERATURE REVIEW

The literature-review section is focused on prior studies concerning merging-behavior modeling using the game-theory approaches and their main issues. Gipps [11] was one of the first to develop a lane-changing model, based on some fixed rules. Later, utility-based models were developed, which considered a binary-choice problem, where the drivers decide whether or not to accept and move to the adjacent gap using a utility-based choice model [12], [13]. Schakel et al. [3] developed regression-based car-following models of the subject vehicle for the gap acceptance-based lane-changing model. However, in many cases, the interaction between the subject vehicle and the following vehicle on the target lane is also important, especially the mandatory lane-changing with merging case.

A game theory-based model can describe such interactions by incorporating the decisions of two drivers in a competing situation where the decisions of drivers depend on the actions of one another.

Kita [14] developed the very first game theory-based merging model. This model considers the interaction as a non-zero-sum, non-cooperative game, which is a common type of game where both players do not cooperate, and their aggregate gains and losses can be less than or more than zero. In this model, time-to-collision (TTC) is the mutual payoff of both the merging and lag vehicles on the mainline. The strategies of the merging vehicle are merging and waiting, while the strategies of lag vehicle are giving way or not. The Nash Equilibrium is utilized to solve the game.

Many studies have made further efforts to improve the payoff formulation of drivers, on this basis. Lopez et al. [15] also adopted TTC, but they compared TTC with several other critical time gaps and formulated the payoff function using the difference between TTC and those critical gaps. Liao et al. [4] combined TTC and headway to form a risk indicator. Liu et al. [16] were the first to consider different objectives for the merging and following vehicles. In this model, the objective of the merging vehicle is to minimize the time spent in the acceleration lane and the objective of lag vehicle is to minimize speed variations. The payoffs are calculated with the expected acceleration depending on the objectives. Peng et al. [17] considered minimizing travel time, Talebpour et al. [18] considered achieving higher speed and avoiding collisions. Then more comprehensive objective and payoff formulations were proposed, including spacing, relative speed, various safety indicators such as time difference to collision (TDTC), change in acceleration, etc. [19], [20], [21], [22].

In terms of action definition, Ali et al. [8] and Chen et al. [9] further defined the following three types of lag vehicle strategies: accelerate, decelerate, and do nothing, for a complete and appropriate consideration. However, the mainline vehicle is also engaged in car-following, and its acceleration behavior is influenced both by the leading vehicle in addition to the merging vehicle. Liao et al. [4] and Min et al. [5] adopted a different game formulation for autonomous vehicles where two players decide who will become the leader.

In the above studies, Liu et al. [16], Talebpour et al. [18], Kang and Rakha [19], Ali et al. [8] and Chen et al. [9] calibrated their model with field-observed data or driving simulator data. Ali et al. [8] utilized the same next generation simulation (NGSIM) and driving simulator dataset to calibrate Liu and Talebpour model. Models were validated and compared, showing that the Ali model outperformed other models in predicting the decision of both the lag vehicle (LV) and the merging vehicle (MV).

Other types of game formulations are also utilized for merging modeling. Kang and Rakha [23] introduced repeated games, considering the merging process as a series of decisions in 1-second interval. The same dataset was used in model validation as Kang and Rakha [19], indicating better performance than their one-shot game. Yu et al. [24] introduced the Stackelberg game where the lag vehicle makes its decision after merging vehicle does. The model was used as autonomous vehicle control and tested in simulation. Lin et al. [21] introduced transferable utility under the connected environments. Numerical simulation was conducted to evaluate the impact of value of time. Arbis and Dixit [25] and Lin et al. [26] employed the quantal-response equilibrium, where driver anticipation of the payoffs is probabilistic in the quantal-response equilibrium to reflect real-driving behavior which has errors in perception. Ji and Levinson [27] compared the game-decision result of user optimization (Nash Equilibrium) and social optimization (Pareto optimum).

These studies mentioned above mainly focus on the lane-changing decisions (deciding the timing of merging into mainline) and longitudinal maneuver (accelerate/decelerate, etc.). There are also some studies focusing on the later movement during the lane-changing process. For example, Lyu et al. [28] focused on identifying lateral movement in merging process and classified it into four driving modes; Seraj et al. [29] proposed a multilane microscopic traffic model combining car-following and lane-changing maneuvers; Guo et al. [30] incorporated two-dimensional time-to-collision measures and deep reinforcement learning to capture evasive lane-change behavior in safety-critical scenarios.

The aforementioned studies make efforts in various aspects, but there are still some important issues to be discussed.

First, in terms of game formulation, the strategies of lag vehicles are not properly defined. There have been three definitions: give way (yield)/not give way (block) [14], [16], [18], [24], accelerate/decelerate [20], accelerate/do-nothing/decelerate [8], [9], [27]. These definitions assume that the lag vehicle only considers the merging vehicle in the merging process. Among these studies, Ali et al. [8] particularly proposed the assumption that lag vehicle is no longer in any car-following status after the merging decision.

For those studies focusing on autonomous vehicles, it is feasible to assume that vehicles conduct simple actions. However, for human drivers, the lag vehicle is in a car-following state in response to both the leading vehicle on the mainline and the merging vehicle. If the lag-vehicle driver remains unaffected by the merging vehicle, the driver will continue car-following to maintain a desired gap from the leading vehicle rather than drive at a constant speed or acceleration. If affected by the merging vehicle, the lag-vehicle driver will adopt behaviors different from regular car-following and adjust the speed and gap to implement the yield or block decisions. It is important to capture such lag-vehicle behaviors.

Second, with the definition of player strategies, it is necessary to identify the corresponding action of vehicles from the observation data. Merging or not can be directly observed through the position of merging vehicle, but the action of lag vehicle needs to be properly extracted. Subjective approaches (by visual observation of trajectories) were adopted in some early studies [16], [18], potentially causing significant errors. Kang and Rakha [19] adopted gap-based rules: yield is indicated by an increased gap between the following and leading vehicles and no yield by a decreased gap. Ali et al. [8] and Chen et al. [9] adopted a more refined rule by converting the speed time series into linear segments and used the slopes of these segments to determine the action. This method also adopts the steady-state theory by Ozaki [31], considering slopes between (-0.05 g, 0.05 g) as the do-nothing state. As mentioned previously, the unaffected lag vehicle continues car-following rather than holding a steady speed, and yield/not yield are compared with keep following rather than using only the acceleration rate. A new identification rule is necessary.

In summary, in a merging scenario, issues related to car-following need to be addressed to model human drivers more realistically.

III. METHODOLOGY

A. GAME DEFINITION

This study describes a merging process under the traditional environment of a taper-type expressway on-ramp section without a parallel acceleration lane; the merging vehicle cannot freely choose where to change to the mainline and must do so at the ending section of ramp.

The typical merging scene is shown in Fig. 1, with a merging vehicle (MV) on the ramp lane, a preceding vehicle (PV) on the mainline target lane, and a lag vehicle (LV) on the mainline target lane. Lane changing is prohibited on the mainline and LV will not change to another lane. This is a strong assumption, and allowing mainline lane changes would be more general. However, due to lane-changing regulations in our study area (as shown in Section IV), we do not consider mainline lane changing in this study.

In this merging scenario, decision making by MV and LV can be modeled as a game with two players, several strategies and a payoff matrix.

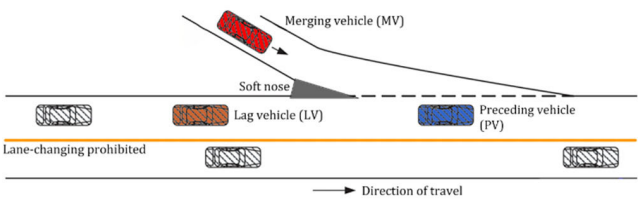


FIGURE 1. Typical merging scenario considered in this study.

This study considers a simultaneous, two-player, non-zero-sum, non-cooperative game. A simultaneous game means that the players make decisions at the same time, a non-zero-sum,

non-cooperative game means players do not cooperate, and their aggregate gains and losses can be less than or more than zero. Player 1 (MV) has two strategies: *merge/wait*, while Player 2 (LV) has three strategies: *yield/keep following/block*. Thus, there are 6 strategy sets: {*merge-yield, merge-keep following, merge-block, wait-yield, wait-keep following, wait-block*}. The definition of vehicle strategy is listed in Table 1 below.

Although Kang and Rakha [17] adopted *overtake* (i.e., MV surpasses the mainline PV) as one of MV strategies, overtaking is not considered in this study because it is very rare in high-density traffic.

TABLE 1. Vehicle strategies.

Player	Strategy	Definition
MV	Merge	Merge into the current gap between LV and PV
	Wait	Wait until LV becomes the new PV
LV	Yield	Give way to current MV by making a larger spacing or decelerating
	Keep following	Don't consider the presence of MV and keep following the current PV
	Block	Try to prevent MV from merging by making a smaller spacing or accelerating

In the game, each player chooses one of the strategies to achieve an equilibrium. In this study, we consider the Nash Equilibrium. The Nash equilibrium is a solution where none of the players has anything to gain by changing only their own strategy. For the game shown in Table 1, in which the strategy of MV is $\boldsymbol{M} = (M_1, M_2)$ and the strategy of LV is $\boldsymbol{L} = (L_1, L_2, L_3)$, the Nash Equilibrium can be expressed as below:

$$\begin{cases} U_1\left(M^*, L^*\right) \geq U_1\left(M, L^*\right) \forall M \in \boldsymbol{M} \\ U_1\left(M^*, L^*\right) \geq U_1\left(M^*, L\right) \forall L \in \boldsymbol{L} \end{cases} \tag{1}$$

where U_1 and U_2 represent the expected payoffs of MV and LV, respectively, and M^* and L^* represent the equilibrium strategy of MV and LV, respectively.

B. PAYOFF FORMULATION

Previous studies supposed various objectives of players and formulated various payoff functions. Note that Ji and D. Levinson [6] argued that different units in the payoff functions of different players could result in a trivial equilibrium solution and thus adopted the required acceleration as a payoff for all strategies, while other studies usually adopted different payoff units. In this study, we considered that different payoff units could describe the objectives of players better and adopted that formulation. Also, we considered that the drivers would adjust speed with different car-following strategies during the merging process, so we did not assume a constant required acceleration rate.

1) REWARD OF MV

It is assumed that MV drivers have different objectives under different strategies. For Player 1 MV, the strategies are *merge* (M_1)/*wait* (M_2).

For the strategy of *merge*, MV is not necessarily assumed to need to increase the speed, and the objective of MV is to avoid collision with LV during merging. For *wait* strategy, the objective of MV is to reduce the time that is wasted when waiting for the current LV to pass, and to ensure merging before reaching the end of the merging section. Thus, the payoff functions of MV consist of time payoff and safety payoff.

C. TIME PAYOFF

t_0 is defined as the expected waiting time before LV sur-passing MV (plus safety distance). MV assumes that LV maintains constant speed and MV maintains an average deceleration rate b_0 when waiting.

$$t_0 = \frac{\Delta v + \sqrt{\Delta v^2 + 2b_0\left(\Delta x + s_0\right)}}{b_0} \tag{2}$$

where

$\Delta v = v_{LV} - v_{MV}\,\Delta x$ is the spacing between two vehicles.

s_0 is minimum accepted spacing between two vehicles.

b_0 is expected deceleration rate when waiting LV to surpass.

D. SAFETY PAYOFF

First, we define I_{risk} as a risk indicator derived from time-to-collision (TTC) and headway.

Only using TTC is not enough to reflect the risk level. Small speed differences result in large times to collision even when the gap is small, and there is still potential danger [30]. So, a hyperbolic tangent function combining TTC and headway is adopted as follows:

$$TTC = \frac{\Delta x}{\Delta v}$$

where

$\Delta v = v_{LV} - v_{MV}\,\Delta x$ is the spacing between two vehicles.

Then, the risk indicator between MV and LV is defined as:

$$I_{risk}^{LV} = \begin{cases} \left(\left[1 - \tanh\left(\frac{TTC}{h_{min}}\right)\right] + \left[1 - \tanh\left(\frac{h}{h_{min}}\right)\right]\right)/2, & TTC \geq 0 \\ \left[1 - \tanh\left(\frac{h}{h_{min}}\right)\right]/2, & TTC < 0 \end{cases} \tag{3}$$

where headway $h = \Delta x / v_{LV}$ and minimum car-following headway h_{min} is obtained from observation data.

For an MV merging before reaching the end of the merging section, the risk indicator is calculated from the projected time to the end of the merging section as follows:

$$I_{risk}^{ramp} = \left[1 - \tanh\left(\frac{t_{ramp}}{h_{min}}\right)\right]/2 \tag{4}$$

where $t_{ramp} = d_{ramp}/v_{MV}$, and d_{ramp} is the ramaining distance to the end of merging area where the width of ramp lane is less than 2.5m and MV must change to mainline.

Following these payoff items, the respective reward functions of MV (p^0) are defined as follows:

$$P_1 = \beta^0_{MV} + \beta^1_{MV} I^{LV}_{risk} + \epsilon_1 \tag{5}$$

$$P_2 = \beta^2_{MV} t_0 + \beta^3_{MV} I^{ramp}_{risk} + \epsilon_2 \tag{6}$$

Finally, $\epsilon_i \sim N(0, 1)$ is the error term, which is different in each payoff function, and β is the coefficient to be calibrated, measuring the importance of the payoff item.

1) REWARD OF LV

For LV, the actions are {*yield/keep following/block*}. LV is assumed to have different payoff formulations in response to MV's action merge or wait.

In the situation where MV chooses *merge*, it is assumed that the objective of LV is to avoid collision with MV and to keep car-following to follow the mainline PV. When MV chooses *wait*, the objective of LV is only to keep following PV.

E. CAR-FOLLOWING KEEPING PAYOFF

Since drivers cannot predict the behavior of other vehicles, a virtual scenario where MV follows the mainline PV is supposed, and LV must compare the result of either following the PV as usual or following with MV. Here we define D_f as the deviation item to evaluate the payoff for LV keeping car-following. Expected trajectories tr_i ($i = 1, 2, 3$) defined below are calculated to measure this deviation at the decision time. In this study, the intelligent driver model (IDM) by Treiber et al. [33] is adopted as the car-following model on mainline lanes. Then, the mean average percentage error (MAPE) is adopted to measure the deviation. The larger the MAPE, the greater the deviation from the original following-leading-vehicle state when interacting with merging vehicles.

Calculating deviation from expected car-following

1.MV assumes that the PV keeps constant speed
2.Calculate expected IDM trajectory $\mathbf{tr_1}$ of MV following PV
3.Calculate expected IDM trajectory $\mathbf{tr_2}$ of LV following the assumed trajectory $\mathbf{tr_1}$ of MV
4.LV assumes to follow PV
5.Calculate expected IDM trajectory $\mathbf{tr_3}$ of LV following PV
6. $D_f = MAPE(\mathbf{tr_2}, \mathbf{tr_3})$

$\mathbf{tr_1}$ represents the situation where MV merges into the mainline without considering LV, $\mathbf{tr_2}$ represents the situation with greatest impact of yielding to MV, and $\mathbf{tr_3}$ represents the situation where LV is not influenced by MV.

Finally, the deviation item is:

$$D_f = MAPE(\boldsymbol{tr_2}, \boldsymbol{tr_3}) \tag{7}$$

When following PV, the relative speed between LV and PV is considered as the payoff:

$$\Delta v_L : v_{LV} - v_{lead} \tag{8}$$

F. SAFETY PAYOFF

I_{risk} of LV is defined to be the same as in (3).

Then payoff function Q_1 when MV chooses *merge* (M_1) and functions Q_2 with MV choosing *wait* (M_2) are defined as below:

$$Q_1 = \beta^0_{LV} + \beta^1_{LV} D_f + \beta^2_{LV} I^{LV}_{risk} + \epsilon_3 \tag{9}$$

$$Q_2 = \beta^3_{LV} + \beta^4_{LV} \Delta v_L + \epsilon_4 \tag{10}$$

where $\epsilon_i \sim N(0, 1)$ is the error term, which is different in each payoff function, and β is the coefficient to be calibrated, measuring the importance of the payoff item.

The payoff of MV and LV under all 6 strategy sets are shown in Table 2 as below:

TABLE 2. Game formulation.

			LV		
			Yield (L_1)	Keep following (L_2)	Block (L_3)
		Probability	p^1_{LV}	p^2_{LV}	p^3_{LV}
MV	Merge (M_1)	p^1_{MV}	(P_{11}, Q_{11})	(P_{12}, Q_{12})	(P_{13}, Q_{13})
	Wait (M_2)	p^2_{MV}	(P_{21}, Q_{21})	(P_{22}, Q_{22})	(P_{23}, Q_{23})

Note that it is also expected that agents place different weights under different situations, thus β_i is different in different joint actions.

G. CAR-FOLLOWING MODEL IN MERGING PROCESS

In this study, longitudinal motion of vehicles is modeled.

As mentioned in Section I, a number of recent studies have simultaneously modeled longitudinal and lateral motions during lane changes. However, this study focuses on the decision-making stage of merging lane changes and on their impact on longitudinal traffic flow characteristics including speed adaptation, and car-following. For this reason, we model only the longitudinal motion and represent the lane-change action as a discrete decision variable, without explicitly modeling the lateral trajectory.

Previous studies assumed that LV was no longer in any car-following mode after the merge decision, while this study considered the car-following behavior to always exist during the merging process, as mentioned above. That is, MV and LV that are engaged in a merging interaction will make decision and follow different car-following strategies to continuously adjust their acceleration and thus speed to achieve the desired gap in merging. This assumption not only keeps the consistency of driving behavior but also enables integration of the

merging model into car-following models and microsimulation. So, this subsection will propose car-following models during the merging process.

Here a modified IDM model is adopted. The original IDM model defines the desired acceleration as below:

$$\frac{dv}{dt} = a[1 - \left(\frac{v}{v_0}\right)^4 - \left(\frac{s^*(v, \Delta v)}{s}\right)^2] \tag{11}$$

where

$$s^* = s_0 + (0, vT + \frac{v\Delta v}{2\sqrt{ab}}) \tag{12}$$

a is maximum acceleration rate;
b is comfortable braking rate;
v_0 is desired speed;
s_0 is minimum standstill distance; and
T is desired time headway.

In IDM, (11) describes that the desired acceleration is decided by the proportion of actual/desired speed and spacing: The closer the actual speed/spacing is to the desired speed/spacing, the smaller the acceleration. And (12) describes that the desired dynamic spacing is decided by both speed and acceleration/braking capability.

Note that the proportion of desired dynamic spacing term s^* and actual distance dominates the braking term. In the original IDM, only the spacing to the preceding vehicle on the same lane is considered. In order to reflect the impact of MV on the adjacent lane, a combination of the spacing to both MV and PV is adopted for LV.

$$s^*_{LV} = \sqrt{ks^*_1 + (1-k)\, s^*_2} \tag{13}$$

$$s_{LV} = \sqrt{ks_1 + (1-k)\, s_2} \tag{14}$$

where

s^*_1 and s_1 are desired/actual spacing to MV;
s^*_2 and s_2 are desired/actual spacing to PV; and
k is the weight parameter to be calibrated.

Other parameters keep the same definition as the IDM. It is assumed that except MV in its wait strategy (keeps a constant deceleration rate and wait), drivers follow different car-following models in other strategies during the merging process. The calibration method and results are presented in the next section.

IV. OBSERVATION DATA AND PROCESSING

A. DATA OVERVIEW

In this study, Zen Traffic Data (ZTD) by Hanshin Expressway was utilized for model calibration and validation. The data was collected along a section of Hanshin Expressway Ikeda Line, in Osaka City, Osaka Prefecture, during September 2018. The observation site is a ~2.1 km two-lane expressway section with an on-ramp entrance (Tsukamoto Entrance) in the middle. The eastward direction (bound for Osaka) was observed. The site map is shown on Fig. 2.

The ZTD contains trajectory data developed from the image sensing technology and the loop detectors. Each trajectory contains the vehicle ID, longitudinal position (kilo-post),

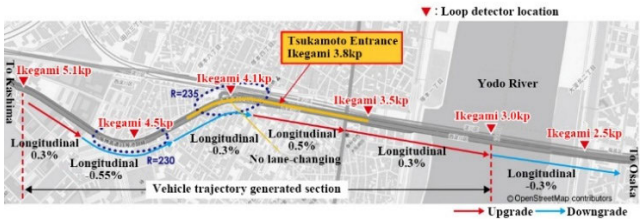


FIGURE 2. Site map of the study area.

lane label (inner lane/outer lane/ramp lane), length of vehicle, and type of vehicle. The observation time interval is 0.1 s.

Around the merging point, the section that is 300 m upstream and 400 m downstream prohibits lane changing, so the simplified scenario is similar to Fig. 1 with a single-lane main-line and a single lane on-ramp section.

B. DATA PREPROCESS

There are five 1-hour datasets available in ZTD. In this study, the morning peak-hour datasets, 7:00-8:00 am and 7:30-8:30 am, were utilized.

ZTD data provides a large quantity of trajectory data, but the position data contains significant noise; thus, the speed and acceleration directly calculated from position data is not reliable for model calibration. In this study, a practical but uncomplicated method is adopted to filter and reconstruct usable trajectories.

Step 1: Outlier removal

In this step, the outliers with unreliable values of acceleration were removed by locally filtering the trajectory. The threshold to detect outliers should be set appropriately high to capture only the measurement errors, and not the random disturbances that affect the observations. A threshold of 30 m/s^2 was adopted as suggested by Montiano and Punzo [34] in reconstruction of NGSIM data. The position points with acceleration beyond this threshold were replaced by natural cubic-spline interpolation with 10 reference points (1)-s data points) before and after the outliers. The average number of outliers was 242 for a trajectory, while the average length of the whole trajectory was 1656 data points.

Step 2: High-frequency noise filtering

In this step, Kalman Filtering was applied to the trajectory in order to filter high-frequency noise. Speed and acceleration were calculated from the processed position, then Kalman Filtering was applied to position, speed, and acceleration time series. The transition matrix of Kalman Filtering was simply set as:

$$F = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

and the filter result was found to be acceptable.

C. ACTION IDENTIFICATION

To get the decision result from observation data, it was necessary to identify the action of both MV and LV. The action

of MV can be directly observed from the relative position of MV and surrounding vehicles through trajectory data. If MV kept to a point in between the initial PV and LV until merging into target lane, the action of MV was *merge*. If the initial LV passed MV and became the new PV before MV merged into target lane, the action of MV was *wait*. However, the decision of LV cannot be directly observed from trajectory data. Previous studies adopt rule-based methods utilizing acceleration, speed, or gap. As mentioned before, this study considered that LV chooses to yield or block by comparing the initial car-following status. A new rule-based method incorporating car-following behavior was adopted to identify the action of LV.

1) CALIBRATION OF IDM PARAMETERS

In order to precisely calculate the regular car-following trajectories, the IDM car-following (CF) parameters were calibrated using the workflow in Fig. 3. In this workflow, calculation of IDM trajectory and measure of performance (MoP) are described below:

Step 1: For each time step t of vehicle i, acceleration was predicted using observed inputs and the car-following (CF) parameter combination.

Step 2: The speed and position were calculated for the next time step. This is actually numerical integration of differential equations. Here the Euler method was adopted.

$$v(t+1) = v(t) + a(t)dt \tag{15}$$

$$x(t+1) = x(t) + v(t+1)dt \tag{16}$$

Step 3: Steps 1 and 2 were repeated to calculate the complete car-following trajectory of vehicle i, and the MoP was calculated from a combination of normalized root-mean-square deviation (NRMSE) of speed and spacing as follows:

$$MoP = \sqrt{\left(\sum \left(v_{predicted} - v_{observed}\right)^2\right)/n} + 0.5* \sqrt{\left(\sum \left(s_{predicted} - s_{observed}\right)^2\right)/n} \tag{17}$$

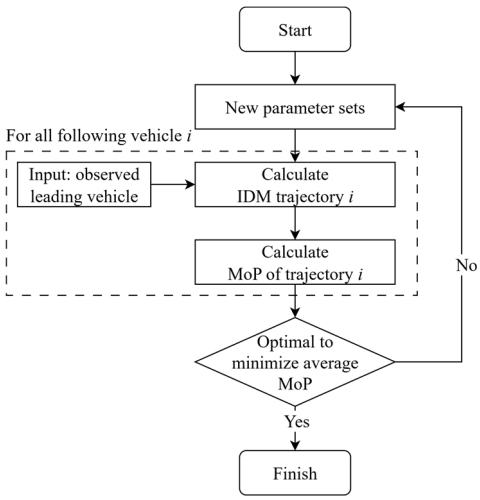


FIGURE 3. IDM parameters calibration workflow.

where v is the speed of following vehicle, s is gross spacing to PV, and n is number of time steps in the trajectory.

Step 4: Steps 1-3 were repeated for all vehicle trajectories i, and the average MoP of all vehicles was calculated as the fitness f of current CF parameter combination.

As shown in (11) and (12), there are five car-following parameters in IDM to be calibrated. The differential evolution (DE) algorithm was adopted to solve the optimization problem of the car-following parameters.

Different form Genetic algorithms (GA), the idea behind the method of DE is that the difference between two vectors yields a difference vector which can be used with a scaling factor to traverse the search space, making it more efficient to search the solutions. Thus, DE are usually reported to outperform GA. To determine the proper method, we randomly select 30 trajectories from the data set and conduct GA and DE to calibrate the car-following parameters with the same population size NP=50. Fitness over iteration is shown in Fig. 4 below, where DE basically converges after 32 iterations, and the result of GA shows oscillation. So, we adopt DE for further calibration.

There are 5 parameters to be calibrated, i.e., number of dimensions $D = 5$, thus the population size is set as $NP = 10D = 50$, crossover probability $CR = 0.6$, and differential weight $F = 0.7$. The number of iterations is set as 200.

The calibrated results are listed in Table 3.

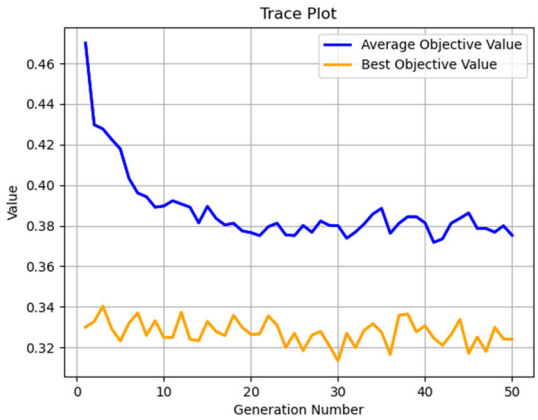
TABLE 3. IDM parameters for mainline car-following.

Parameter	Calibrated value	Typical value
Desired speed v_0 (m/s)	22.52	Speed limit is 16.67 on the Hanshin expressway
Standstill distance s_0 (m)	2.55	2.00
Minimum headway t_0 (s)	1.15	1.60
Maximum acceleration a (m/s^2)	0.74	0.73
Comfortable brake rate b (m/s^2)	2.33	1.67

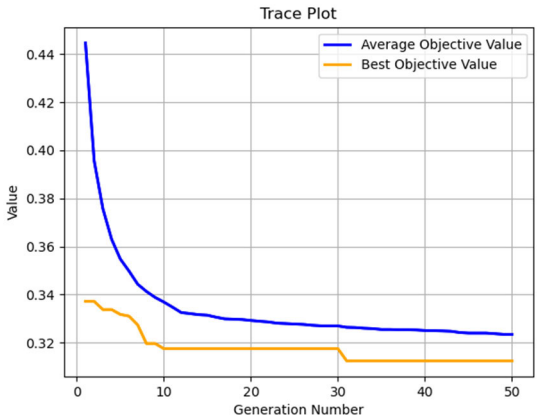
The drivers on Hanshin Expressway tended to drive over the speed limit and showed a more aggressive behavior with shorter headway and higher acceptable braking rate compared with IDM typical value suggested by [31]. Although results also show a larger standstill distance, spacing will be shorter when the cars are moving due to significantly shorter headway. The spacing under IDM reference parameters is $s_1 = 2.0 + 1.6v$, and $s_2 = 2.55 + 1.15v$ under calibrated parameters. It's obvious that when speed is higher than 1.22 m/s (4.4 km/h), the actual spacing will be shorter in the studied situation, indicating a more risky driving.

2) RULE-BASED METHOD

We used the concept of deviation from expected car-following trajectory for judgment, which is similar to that in Section III. The rules are listed below.



(a) GA



(b) DE

FIGURE 4. **Comparison of GA and DE in fitness over iteration.**

First, we supposed that LV keeps following the PV and used IDM to calculate the expected trajectory for LV, using the calibrated car-following parameters. Then we calculated the MAPE against observed and expected trajectory of LV. The distribution of MAPE is shown in Fig. 5 below. It can be found that it shows a 2-peaked distribution, and the transition point is about 0.12. We can infer that the two peaks may represent two different statuses of LV: keeping following, or interrupted by MV. Thus, we adopted 0.12 as the threshold to determine action of LV.

a) If the MAPE against observed speed and IDM-predicted speed were under 0.12 threshold (indicating that the observed trajectory was similar enough to ideal car-following behavior), it was identified as *keep following*.
b) If the MAPE was over 0.12 threshold, and the observed spacing between LV and PV was smaller than IDM-predicted gap at the end of merging process (indicating that there is a reduced gap compared to regular following), it was identified as *block*.
c) If the MAPE was over 0.12 threshold, and the observed spacing was larger than the IDM-predicted gap (indicating an increased gap compared to regular following), it was identified as *yield*.

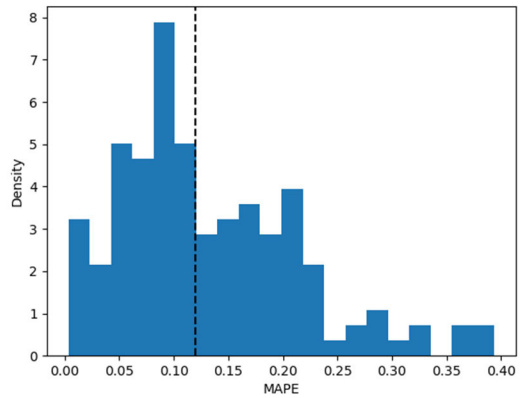


FIGURE 5. **Distribution of deviation (MAPE) against observed and expected trajectory of LV.**

3) ACTION IDENTIFICATION RESULT

We then identified the actions of LV by substituting the IDM parameters into the rules mentioned above. Fig. 6 illustrates the three types of LV actions.

The results shown in Table 4 combine the identified actions of both MV and LV.

TABLE 4. **Summary of vehicle actions.**

Action type	N of observations	Percentage
Merge-Yield	177	48.76%
Merge-Keep following	134	36.91%
Merge-Block	10	2.75%
Wait-Yield	15	4.13%
Wait-Keep following	20	5.51%
Wait-Block	7	1.93%
MV Merge	311	85.67%
MV Wait	52	14.33%
LV Yield	192	52.89%
LV Keep following	154	42.42%
LV Block	17	4.68%

86% of MV succeed to merge into the mainline without waiting, while only 5% of LV chose to take blocking action, suggesting that although drivers tend to drive more aggressively on the mainline, they still need to give way to ramp vehicles. It is also worth mentioning that no illegal lane-changing to inner lane was observed in MV trajectories, which is also consistent with the assumption of not considering overtaking as mentioned in Section III.

If we compare the action-identification results of NGSIM data from previous studies—despite different identification methods and different samples—these studies report a much higher proportion of wait for MV: 76.1% of 2389 in [8], and 53.8% of 429 in [9] and also a lower proportion of yield for LV: 10.1% of 2389 in [8], and 12.1% of 429 in [9]. This indicates that the short length of the merging section and

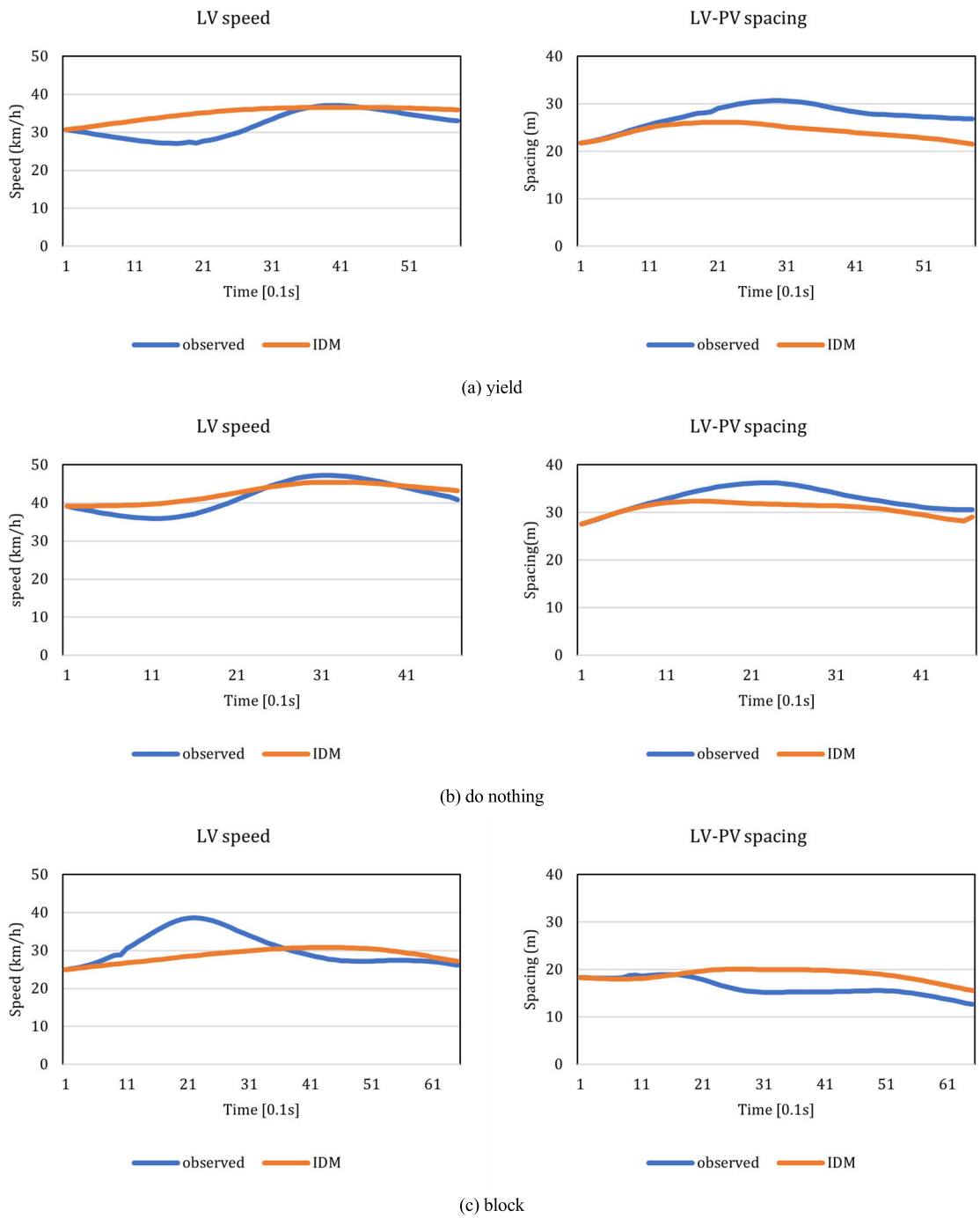


FIGURE 6. **An example of speed and spacing profile of LV action.**

absence of acceleration lane do have an important impact on the merging interaction.

V. MODEL CALIBRATION AND VALIDATION

A. CALIBRATION OF CAR-FOLLOWING MODEL

As mentioned in Section III, the modified IDM model was proposed to reflect the car-following behavior during merging. The calibration method is basically the same as described in Section IV but utilized the trajectories of a lead-MV-LV group instead of a lead-follow pair. In order to simplify calculations, the desired speed v_0, maximum acceleration rate a, and comfortable brake rate b, are set as the same as those of the mainline vehicles.

The calibrated results are shown in Table 5.

We found that during the merging process, drivers under all three strategies tended to keep a shorter standstill distance.

TABLE 5. Car-following parameters for different strategies.

Parameter	Mainline	LV *yield*	LV *block*	MV *merge*	MV *wait*
Desired speed v_0 (m/s)		22.52			
Standstill distance s_0 (m)	2.55	1.63	1.51	1.19	
Minimum headway t_0 (s)	1.15	2.09	0.41	0.62	
Maximum acceleration a (m/s^2)		0.74			
Comfortable brake rate b (m/s^2)		2.33			
Weight of spacing k	-	0.34	0.37	-	
Waiting brake rate (m/s^2)					-0.4

A yielding LV tends to keep a longer headway to giving a spacing, while a blocking LV or MV tended to keep a much shorter headway to reduce the spacing (which is empirically reasonable).

B. CALIBRATION OF GAME MODEL

This study adopted the bi-level programming method proposed by Kita [14] to estimate the game model parameters. The upper-level programming was a non-linear programming problem, to minimize the error between the observed and the predicted actions.

Since the sample is imbalanced with very few cases MV's wait action and LV's block action, we adopted a weighted objective function to balance the sample. The weight for each action type is:

$$w_i = 1/\sqrt{n_j} \tag{18}$$

$$\min \sum_{i=1}^{n} w_i(1 - g_i) \tag{19}$$

where

$n_j (j = 1, 2, \ldots, 6)$ is the number of observations of each action type (Merge-Yield, Wait-Block, etc.). n is the number of observations.

g is error indicator, $g = 1$ if observed action combination (*merge*/*wait* for an MV, *yield*/*keep following*/*block* for an LV) is the same as predicted action, $g = 0$ if not.

The calibration result with balanced weight was compared with unbalanced weight:

$$\min \sum_{i=1}^{n} (1 - g_i) \tag{20}$$

The lower-level programming seeks the solution for the Nash equilibrium. To solve the two player non-cooperative game, *nashpy* package in Python was utilized in this study to obtain Nash equilibrium. The whole calibration framework is shown on Fig. 7.

DE algorithm is also used in this study to solve the minimize problem in this section. Parameters of DE are shown as follows:

The are 22 parameters to be calibrated, and population size is set as $NP = 5D = 110$, crossover probability $CR = 0.6$, and differential weight $F = 0.7$. The number of iterations is set as 500. In order to avoid over fitting and improve efficiency, 75 trajectories are randomly selected for calculation in each iteration. 254 out of 363 (70%) observations were randomly selected for calibration, and the remaining data was used for cross validation.

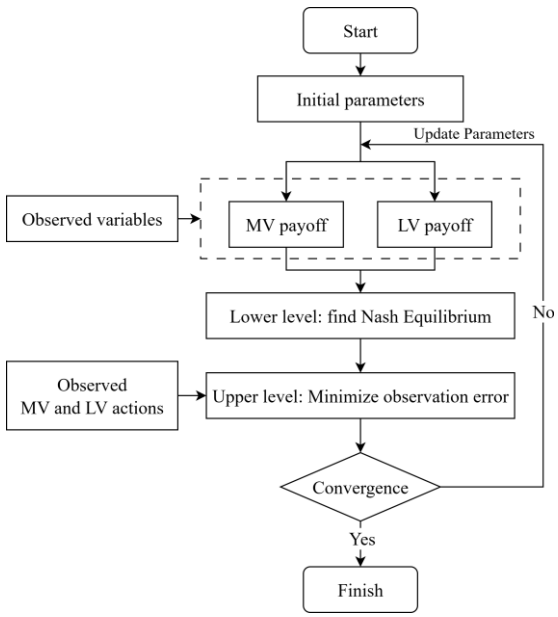


FIGURE 7. Game-model calibration framework.

C. MODEL VALIDATION

This section presents the prediction capability of the calibrated model.

1) VALIDATION OF PREDICTION RATE AND COMPARISON WITH OTHER MODELS

The remaining 109 (30%) observations were used for cross validation, and the prediction rate for each strategy (merge/wait for MV, yield/keep following/block for LV), respectively.

First, calibration result with balanced weight was compared with unbalanced weight. The result in Table 6 indicates that using objective function with balanced weight can improve the prediction rate of minority cases without reducing the rate of majority cases. Therefore, the result of balanced weight was adopted as shown in Table 7.

We found that the model could successfully predict 97.9% of MV behavior and performed well for the *merge*strategy. The model could also predict 72.5% of LV behavior, the performance for *yield* and *keep following* was acceptable while the performance for *block* was limited because of the small sample size for *block*, but balanced weight in objective function still improved the performance.

TABLE 6. Validation of balanced weight.

		Unbalanced		Balanced	
	N of obs	Correct prediction	Prediction rate	Correct prediction	Prediction rate
MV-Merge	97	95	0.979	95	0.979
MV-Wait	12	8	0.667	9	0.750
MV-overall	109	103	0.945	104	0.954
LV-Yield	59	41	0.694	40	0.678
LV-Keep following	45	35	0.778	36	0.800
LV-Block	5	2	0.400	3	0.600
LV-overall	109	77	0.706	79	0.725

TABLE 7. Reward function parameters.

Player	Joint action	Const $\boldsymbol{\beta^0_{MV}}$	Risk index (I^{LV}_{risk}) $\boldsymbol{\beta^1_{MV}}$	
MV	Merge-Yield	-0.37	-1.43	
	Merge-Keep following	3.40	-4.69	
	Merge-Block	-2.75	1.04	
		Const	Time wasted (t_0) $\boldsymbol{\beta^2_{MV}}$	Risk index (I^{ramp}_{risk}) $\boldsymbol{\beta^3_{MV}}$
	Wait-Yield	-	-1.86	-0.62
	Wait-Keep following	-	3.71	-3.68
	Wait-Block	-	-0.69	-0.90
		Const $\boldsymbol{\beta^0_{LV}}$	Deviation from following (D_f) $\boldsymbol{\beta^1_{LV}}$	Risk index (I^{LV}_{risk}) $\boldsymbol{\beta^2_{LV}}$
LV	Merge-Yield	-0.53	2.38	4.88
	Merge-Keep following	-	-	*Utility* = 0
	Merge-Block	-3.05	-0.49	-4.30
		Const $\boldsymbol{\beta^3_{LV}}$	Speed difference (Δv_L) $\boldsymbol{\beta^4_{LV}}$	
	Wait-Yield	-4.94	-3.80	
	Wait-Keep following	-	-	*Utility* = 0
	Wait-Block	-4.87	3.51	

For comparison, the model proposed by Ali et al. [8] was also calibrated and validated with the same setup in this study. Since the strategy definition and identification methods were different, only the prediction capability of MV was compared, and the result is shown in Table 6 below. As for reference, with NGSIM data, Ali's study reported an 88% prediction rate for MVs and 51% for LVs, Chen et al. [9] reported 81% for MVs without validating LVs. The calibration and comparison results are shown in Table 8.

Then the variance-based Sobol' method [35] was adopted to for global sensitivity analysis and measure the effects of calibrated parameter. The predicted probability for MV to choose *merge* action is selected as output, and the variation of parameters is set as $\pm 10\%$. A sample sequence of 1000 points inside the unit hypercube is sampled using Monte Carlo method for each merging trajectory. The first-order indices of parameters are shown in Table 9.

TABLE 8. Model comparison.

		Proposed Model		Reference model	
	N of obs	Correct prediction	Prediction rate	Correct prediction	Prediction rate
MV-Merge	97	95	0.979	93	0.959
MV-Wait	12	9	0.750	7	0.583
MV-overall	109	104	0.954	100	0.917
LV-Yield	59	40	0.678		
LV-Keep following	45	36	0.800		
LV-Block	5	3	0.6		
LV-overall	109	79	0.725		

TABLE 9. Sensitivity analysis.

Player	Strategy	Const	Risk index (I^{LV}_{risk})	
MV	Merge-Yield	0.04	0.21	
	Merge-Keep following	**0.50**	0.02	
	Merge-Block	0.31	0.24	
		Const	Time wasted (t_0)	Risk index (I^{ramp}_{risk})
	Wait-Yield	-	**0.49**	0.12
	Wait-Keep following	-	0.06	0.09
	Wait-Block	-	0.27	0.05
		Const	Deviation from following ($MAPE$)	Risk index (I^{LV}_{risk})
LV	Merge-Yield	0.19	0.06	**0.33**
	Merge-Keep following	-	-	*Utility* = 0
	Merge-block	0.13	0.14	**0.40**
		Const	Speed difference (Δv)	
	Wait-Yield	0.27	0.01	
	Wait-Keep following	-	-	*Utility* = 0
	Wait-Block	0.06	0.18	

The results suggest that parameters related to drivers' perception of time loss and risk contribute most to the variation in merging decisions. For example, the constant term $\boldsymbol{\beta^0_{MV}}$ in the payoff function of the MV under the merge–keep-following strategy and the time-wasted coefficient $\boldsymbol{\beta^2_{MV}}$ under the wait–yield strategy obtained the highest first-order indices, suggesting that drivers' intrinsic preference and aversion to delay strongly influence their choice to merge or wait. For the LV, the risk indicator $\boldsymbol{\beta^2_{LV}}$ in the merge–block scenario exhibited the largest sensitivity, indicating that blocking

behavior is not purely aggressive but rather constrained by the driver's safety perception. These findings imply that human merging behavior is influenced not only by physical vehicle dynamics but also by subjective assessments of safety and efficiency. From a practical perspective, this suggests that autonomous vehicle controllers need to account for human drivers' trade-offs between time and risk.

2) VALIDATION OF CAR-FOLLOWING MODEL

This study incorporated merging-decision games and car-following model, so the capability of trajectory prediction was also examined.

For each observed trajectory of a PV-MV-LV group, the strategy decision of MV and LV were predicted with the game model, and then their trajectories were predicted with corresponding car-following models proposed in this study. The MAPE against predicted and observed trajectories was utilized to measure the prediction error.

The MAPE for MV was 0.124, and that of LV was 0.105, indicating a reasonable trajectory prediction capability.

VI. CONCLUSION

This paper describes a game theory-based lane-changing model incorporating car-following behavior for merging-section modeling. The proposed model extends existing models by considering the keep following strategy of LV. In this model, the merging decision-making behavior was modeled as a two-player, non-zero-sum, noncooperative game with complete information. To formulate the payoffs of MV and LV, safety, travel time, and speed deviation from following-PV were selected as key factors. Based on the ZTD of the Hanshin Expressway, a rule-based method was proposed to identify the vehicle actions from observation. Models were calibrated with extracted data. The validation section indicated that the proposed model has reasonable capability in both decision-making prediction and trajectory prediction. There are a number of taper-type merging sections on the urban expressways in Japan. As reported by police agency [36], in 2024, 107 out of 4033 (2.65%) accidents happened on ramp areas of arterial expressways, while 57 out of 1388 (4.11%) accidents happened on ramp areas of urban expressways, suggesting higher risk. The findings in this study can help to predict the merging situation and prevent accidents on those sites.

Considering practical implications, the proposed model can be embedded into microscopic traffic simulation tools to provide more realistic predictions of merging interactions at taper-type ramps where drivers' behaviors are different from the widely studied parallel-type ramps. Such simulations can help to evaluate the effectiveness of infrastructure measures such as ramp length, taper design, or the implementation of ramp metering. Besides, the payoff-based game theory framework could be extended to automated vehicle controllers, such as designing reinforcement learning models, enabling them to anticipate the strategies of human-driven vehicles and make safer and smoother merging decisions.

Since the sample size was not large enough, the block strategy of LV was rarely observed, resulting in a relatively lower prediction rate for LV in the model. Although it is feasible to calibrate the parameters with an objective function with balanced weight, the sample size is still a point to be improved by more observation data. Another limitation is that the lane-changing action of mainline vehicles is not considered. This study provides a validated modeling framework applicable to taper-type merging sections, making it possible to extend the action types. In future, more observation data at different locations will be required. This study proposed modified IDM car-following models to capture the speed-adjustment behavior during merging, so in future the entire merging model can be integrated into microsimulation for further study.

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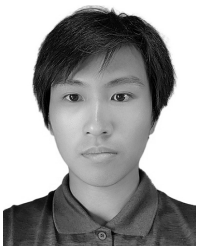

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